

The effect of the fire regulation cover increase on the column confinement zone

TS500 column clear cover is 25mm, and the distance of longitudinal reinforcement to the edge is taken as 40mm. If the clear cover for longitudinal reinforcement is increased to 50mm due to fire regulations, the minimum transverse reinforcement condition related to column confinement zones in TBDY-7.3.4.1 article; When the concrete strength value in Equation 7.1 formula is high or the clear cover increases, the transverse reinforcement ratio increases. As the transverse reinforcement ratio increases, the number of reinforcements must also be increased to place stirrups, and in the longitudinal reinforcement ratio, this indirectly causes an increase in transverse reinforcement.

Fire regulations

Article 3- The fifth paragraph of Article 23 of the same regulation has been amended as follows.

"(5) Relevant regulations and standards are complied with in the structural system elements made of reinforced concrete and prestressed concrete. In multi-storey buildings and especially in buildings with horizontal fire partitions, the system is examined as a whole, taking into account the additional forces arising in cases where element expansions are restricted. In order for reinforced concrete or reinforced concrete-steel composite elements to be resistant to fire for 120 minutes according to Annex-3/B, the net concrete cover size, which is the distance between the outer face of the outermost steel profile or reinforcement and the outermost concrete fiber, must be at least 35 mm in columns, 25 mm in beams and at least 20 mm in slabs. TS500 standard is followed for the structural system elements made of concrete with fire resistance less than 120 minutes.

2018 Earthquake regulations

7.3.4.1 - Special confinement zones will be created at the lower and upper ends of each column.

The length of each confinement zone, measured from the top elevation of the floor upwards or from the bottom surface of the highest beam connected to the column downwards, shall not be less than 1/6 of the column's free height, 1.5 times the column's largest cross-section dimension, and 500 mm. In cantilever columns, the confinement zone will be created at the bottom of the column, and its length shall not be taken less than twice the column's large dimension. The conditions for transverse reinforcement in confinement zones are given in (a) to (d) below.

These reinforcements will continue for a height not less than the minimum column dimension within the foundation.

However, in cup foundations supporting columns, the transverse reinforcement in the confinement zone will continue throughout the cup height.

(a) Transverse reinforcement with a diameter smaller than $\phi 8$ shall not be used in confinement zones.

In this zone, the spacing of longitudinal stirrups and ties shall not be greater than 1/3 of the smallest cross-section dimension, 150 mm, six times the longitudinal reinforcement diameter, and not less than 50 mm. The horizontal distance between stirrup legs and/or ties, a , shall not be greater than 25 times the stirrup diameter. The pitch of continuous circular spirals shall not be greater than 1/5 of the core diameter and 80 mm. In circular columns, all confinement reinforcement shall be provided by circumferential transverse reinforcement.

(b) In columns with stirrups, where $N_d > 0.20 A_c f_{ck}$ (compression) the minimum total transverse reinforcement area in confinement zones shall be calculated to satisfy the less favorable condition of Eq.(7.1).

In this calculation, the core dimension b_k shall be considered separately for both directions (Figure 7.3):

$$\begin{aligned} A_{sh} &\geq 0.30 s b_k [(A_c / A_{ck}) - 1] (f_{ck} / f_{ywk}) \\ A_{sh} &\geq 0.075 s b_k (f_{ck} / f_{ywk}) \end{aligned} \quad (7.1)$$

(c) In columns with circular reinforcement, where $N_d > 0.20 A_c f_{ck}$ (compression), the minimum volumetric ratio of transverse reinforcement in confinement zones shall be calculated to satisfy the less favorable condition of Eq. (7.2).

$$\rho_s \geq 0.45 [(A_c / A_{ck}) - 1] (f_{ck} / f_{ywk})$$

$$\rho_s \geq 0.12 (f_{ck} / f_{ywk}) \quad (7.2)$$

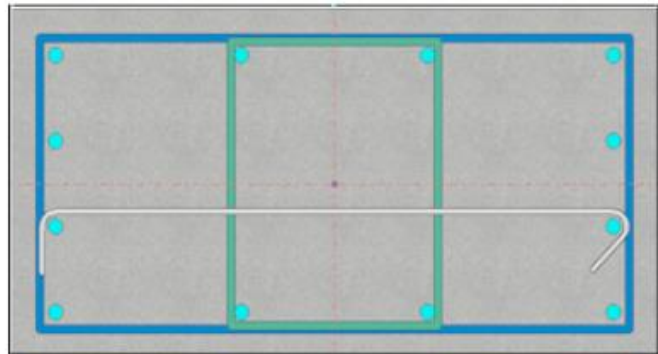
Control of reinforcement density in the confinement zone due to the stirrup in the 30x60cm column:

We will control the reinforcement density based on volumetric density.

b=30cm, d=60cm, d'=4cm as reinforcement density in the confinement zone:

Material: C30/S420, fck=300 kg/cm², fywk=4200 kg/cm²

```
End confined zone rebar check
Asw1= 0.30*s*bk[(Ac/Ack)-1] * (fck/fywk)
Asw2= 0.075*s*bk*(fck/fywk)
Asw = max(Asw1,Asw2)
Nd< 0.2*Ac*fck, Ash > 2/3*Asw
9627< 0.2*1800*300=108000
Stirrup: Ø8/8
X direction check
Ash=3*0.5=1.51 cm²
Asw1=0.30*8*25*[1800/1375-1]*(300/4200)=1.32
Asw2= 0.075*8*25*(300/4200)=1.07
Asw=2/3 max(Asw1,Asw2)=0.87
Ash= 1.51>Asw=0.87
Y direction check
Ash=4*0.5=2.01 cm²
Asw1=0.30*8*55*[1800/1375-1]*(300/4200)=2.91
Asw2= 0.075*8*55*(300/4200)=2.36
Asw=2/3 max(Asw1,Asw2)=2.91
Ash= 2.01>Asw=1.92
```

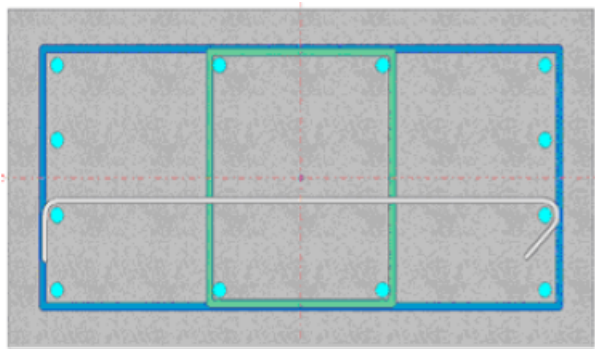


Ac=30x60=1800cm², Ack= (30-2.5) x (60-2.5) = 1375cm²

Ro1=0.30 [Ac/Ack-1] fck/fywk= 0.00662 , Ro2=0.075 fck/fywk= 0.00535

b=30cm, d=60cm, d'=5cm as reinforcement density in the confinement zone:

```
End confined zone rebar check
Asw1= 0.30*s*bk[(Ac/Ack)-1] * (fck/fywk)
Asw2= 0.075*s*bk*(fck/fywk)
Asw = max(Asw1,Asw2)
Nd< 0.2*Ac*fck, Ash > 2/3*Asw
9627< 0.2*1800*300=108000
Stirrup: Ø8/5
X direction check
Ash=3*0.5=1.51 cm²
Asw1=0.30*5*23*[1800/1219-1]*(300/4200)=1.17
Asw2= 0.075*5*23*(300/4200)=0.62
Asw=2/3 max(Asw1,Asw2)=0.78
Ash= 1.51>Asw=0.78
Y direction check
Ash=4*0.5=2.01 cm²
Asw1=0.30*5*53*[1800/1219-1]*(300/4200)=2.71
Asw2= 0.075*5*53*(300/4200)=1.42
Asw=2/3 max(Asw1,Asw2)=2.71
Ash= 2.01>Asw=1.79
```



Ac=30x60=1800cm², Ack=(30-3.5)x(60-3.5)=1219cm²

Ro1=0.30 [Ac/Ack-1] fck/fywk= 0.0102 , Ro2=0.075 fck/fywk= 0.00535

The increase in reinforcement density in the confinement zone due to an increase of account cover 1cm in the calculation stirrup in the column;

0.0102/0.00662 = 1.54 times increase.

When the confinement zone width is $\phi 8/8$, it can be met with $\phi 8/5$ spacing.

In the fire regulation, it is required to take 3.5cm of the outermost concrete cover of the column reinforcement. In a 40mm calculation account cover, the clear cover is 20mm. In the case of taking a clear cover of 35mm, a calculation account cover of 55mm (35+20) is required. In this case, it will be more difficult to meet the reinforcement conditions in the confinement zone.

In detailing options, you can see the difference when the confinement zone options are closed. This condition is also used as a mandatory condition in other regulations such as ACI318, Eurocode, and IBC.

STA4-CAD DETAILING OPTIONS

PLOTTER SETUP | DETAILING OPTIONS | ANALYSIS OPTIONS

DESIGN CODES | SEISMIC CODES | NONLINEAR ANALYSIS

COLUMN DESIGN OPTIONS

Page 1 | Page 2 | Page 3 | Page 4

Column confinement zone horizontal minimum rebar percentage

Rectangular column confinement zone horizontal minimum rebar percentage

☒ Ash > 0.30 s bk fck / fywk [Ac/Ack-1] ☒ Ash > 0.075 s bk fck / fywk

☐ Calculation of polygon column arms like rectangular columns

Circular column confinement zone horizontal minimum rebar percentage

☒ Ash > 0.45 s bk fck / fywk [Ac/Ack-1] ☒ Ash > 0.12 fck / fywk

TBDY 7.3.4.1 It is valid for all columns.

2018 TURKISH BUILDING SEISMIC CODE

Confined column strength multiplier <1.7>

Unconfined column strength multiplier <1.0>

$V_e = C \times b \times h \times \sqrt{f_{ck}}$

Beam-Column Connection Shear effective width

☒ $b_j = \min(Bw1, Bw2, Bc, (Bw1 + 2 \cdot X1), (Bw2 + 2 \cdot X2))$

☐ $b_j = \min(Bw1, Bw2, Bc, (Bw1 + h_c / 2), (Bw2 + h_c / 2))$

☐ $b_j = \min((Bw1 + Bw2) / 2, Bc)$

TDY2007, TBDY 2018, ACI 318

SHEARWALL ENDZONE OPTIONS

☐ Shearwall end zone, 3 stage transition option

☐ End zone percentage $A_s/A_c < 0.03$

TDY 3.3.4.1 It is valid for all

High ductility Cantilever beams option

☐ Cantilever beams will not be counted

☒ Taken into account to M_r capacity

☐ Taken into account to M_d design

☐ All beams calculate according to M_d design

The confinement zone tie spacing options can be adjusted in the column rebar selection section.

STA4-CAD COLUMN TIE OPTIONS

COLUMN INSIDE STIRRUP OPTION

$\max \frac{D}{B} \leq 2$

☒ Columns perform 2.inside stirrup.

$\max \frac{D}{B} \leq 1.25$ Default

COLUMN CONFINED ZONE TIE SPACE OPTION

☒ Consider overlapping longitudinal rebars in the column stirrup space

>> min zone rebar = $\phi 16$ 50 cm

COLUMN CONFINED ZONE

Min. space=e

$\phi 8$ 5

$\phi 10$ 6

Except for the 6 ϕ condition

☒ Structural wall critical zone, end zone stirrup condition

6 ϕ > 8 cm >> min zone rebar = $\phi 14$

☒ Use of welded wire in polygonal shearwall